

Practice Qual

Exercise 1. (January 2020 Problem 1) On this problem, no justification is required; only the answer will be graded. Let S_3 be the symmetric group on 3 letters, and let R be the group ring $\mathbb{Z}[S_3]$.

- (a) Write down a nonzero element of R which is a zero-divisor.
- (b) Write down an element in the center of R which is not in $\mathbb{Z}$.
- (c) Write down an R -module which is not a free module.

Exercise 2.

- (a) Let $E \supset \mathbb{Q}$ be a degree 4 extension, and suppose there are two distinct subextensions $K_1, K_2 \subset E$, $K_1, K_2 \supset \mathbb{Q}$. Prove that $E/\mathbb{Q}$ is Galois.
- (b) Let $E \supset \mathbb{Q}$ be a degree 4 extension, and suppose $\sqrt{-m} \in E$ for some positive $m \in \mathbb{Z}$. Prove that $E/\mathbb{Q}$ is Galois.

Exercise 3. Given a group G and a subgroup H , the **normalizer** of H , denoted $N(H)$, is the set of all elements $g \in G$ so that $g^{-1}Hg = H$. That is, $N(H) = \{g \in G : g^{-1}hg \in H \text{ for all } h \in H\}$.

- (a) Prove that H is a normal subgroup of $N(H)$.
- (b) Let G be a group and H a subgroup. Let H act on the left cosets of H by left multiplication: $h(gH) = hgH$. Prove that gH is a fixed point for this action if and only if $g \in N(H)$.
- (c) Suppose that H has order p^k for some prime p . Prove that if gH is not a fixed point of the action, then the size of its orbit is divisible by p .
- (d) Prove that $N(H)/H$ has size divisible by p .

Exercise 4. Let R be a commutative ring. Recall that a chain of prime ideals of length k is a sequence of $k + 1$ prime ideals, each a proper subset of the next:

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_k.$$

The **dimension** of R , $\dim(R)$, is defined to be the length of the longest chain of prime ideals contained in R .

- (a) Suppose R is a PID. Prove that $\dim(R) = 1$.
- (b) Prove that $\dim(\mathbb{C}[x, y]) \geq 2$.
- (c) Let $S \subset R$ be a multiplicative set. Prove that $\dim(R[S^{-1}]) \leq \dim(R)$.
- (d) Give an example of a ring R and a multiplicative set S so that $R[S^{-1}]$ is not a field and $\dim(R[S^{-1}]) < \dim(R)$.

Exercise 5. (Based on a problem from another school's algebra qual) Let $M_2(\mathbb{C})$ be the space of 2×2 matrices over $\mathbb{C}$. The space $M_2(\mathbb{C})$ is a complex vector space of dimension 4. There is a linear map $T: M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$ that sends A to A^T .

- (a) Compute the eigenvalues of T .
- (b) For each eigenvalue of T , compute a basis for the associated eigenspace.
- (c) Is the map T diagonalizable?